Lecture 19: Strong Duality and Saddle Points

MA3236 Non-Linear Programming

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Quick review: lower bounds and duality gaps

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f(\mathbf{x}) + \boldsymbol{\lambda}^T \mathbf{g}(\mathbf{x}) + \boldsymbol{\nu}^T \mathbf{h}(\mathbf{x}), \qquad \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x} \in \mathbf{R}^n} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}).$$

For primal feasible $\mathbf{x}$ and $(\boldsymbol{\lambda}, \boldsymbol{\nu}) \in \mathcal{F}_D = \operatorname{dom} \theta \cap (\mathbf{R}^m \times \mathbf{R}^p_+)$,

$$\boxed{-\infty < \theta(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq d^* \leq p^* \leq f(\mathbf{x}) < +\infty.}$$

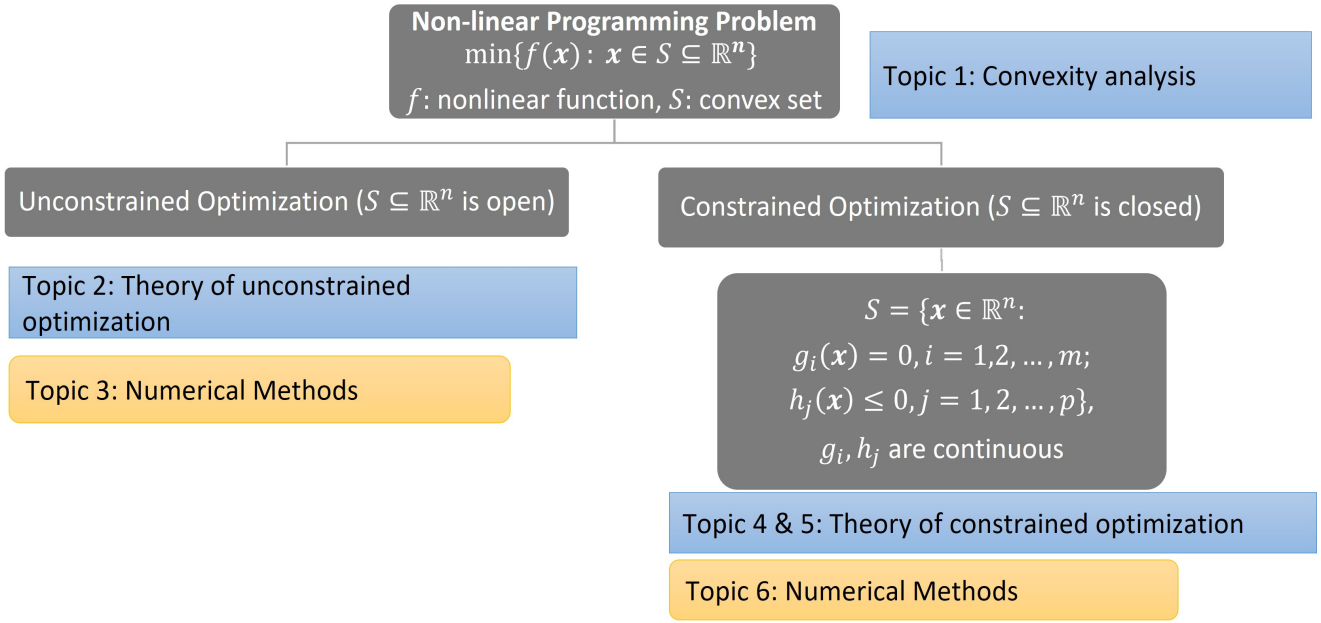
- Objective error bound: $f(\mathbf{x}) - p^* \leq f(\mathbf{x}) - \theta(\boldsymbol{\lambda}, \boldsymbol{\nu})$.
- The optimal duality gap $p^* - d^* \geq 0$ **need not be zero**.

Quick review: what the dual viewpoint gains and costs

$$\sup_{\boldsymbol{\lambda}\in\mathbf{R}^m,\,\boldsymbol{\nu}\geq\mathbf{0}} \underbrace{\inf_{\mathbf{x}\in\mathbf{R}^n} L(\mathbf{x},\boldsymbol{\lambda},\boldsymbol{\nu})}_{\theta(\boldsymbol{\lambda},\boldsymbol{\nu})}.$$

Attractive features	Questions that remain
Convex problem	$\mathcal{F}_D$ can be restrictive
Fewer variables if $m + p < n$	Finiteness, attainability, uniqueness
Unconstrained inner minimization	Global inner solves, nonsmooth dual

Topics



In Topic 5, we study dual formulations and optimality theory for **constrained optimization problems**.

- Lagrangian dual and weak duality theorem.
- Strong duality theorem and saddle point optimality condition.

Convex strong duality

Consider

$$\min_{\mathbf{x}\in\mathbf{R}^n} f(\mathbf{x}) \quad \text{s. t.} \quad \mathbf{B}\mathbf{x} = \mathbf{b}, \quad h_j(\mathbf{x}) \leq 0.$$

Theorem 19.1 (Convex strong duality)

Suppose f, h_j are convex and finite on $\mathbf{R}^n$, $p^* \in \mathbf{R}$, and there exists $\widehat{\mathbf{x}} \in \mathbf{R}^n$ such that

$$\mathbf{B}\widehat{\mathbf{x}} = \mathbf{b}, \qquad h_j(\widehat{\mathbf{x}}) < 0 \quad \text{for every } j.$$

Then $d^* = p^*$, and a dual optimizer $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ exists.

The strict-feasibility assumption is Slater's condition.

NB. The existence of a primal optimizer is not asserted.

Illustration

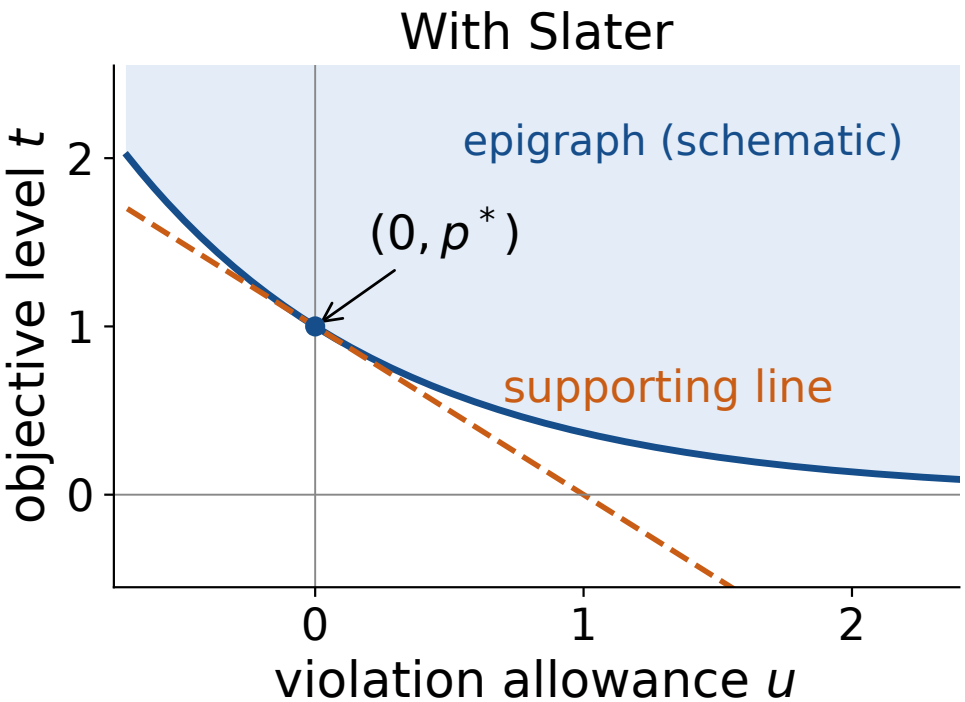
f, h convex $\Longrightarrow$ convex shaded area:

$$\{(u,t) : \exists\, \mathbf{x} \in \mathbf{R}^n, \quad f(\mathbf{x}) \leq t, \quad h(\mathbf{x}) \leq u\}\,.$$

Illustration

f, h convex $\Longrightarrow$ convex shaded area:

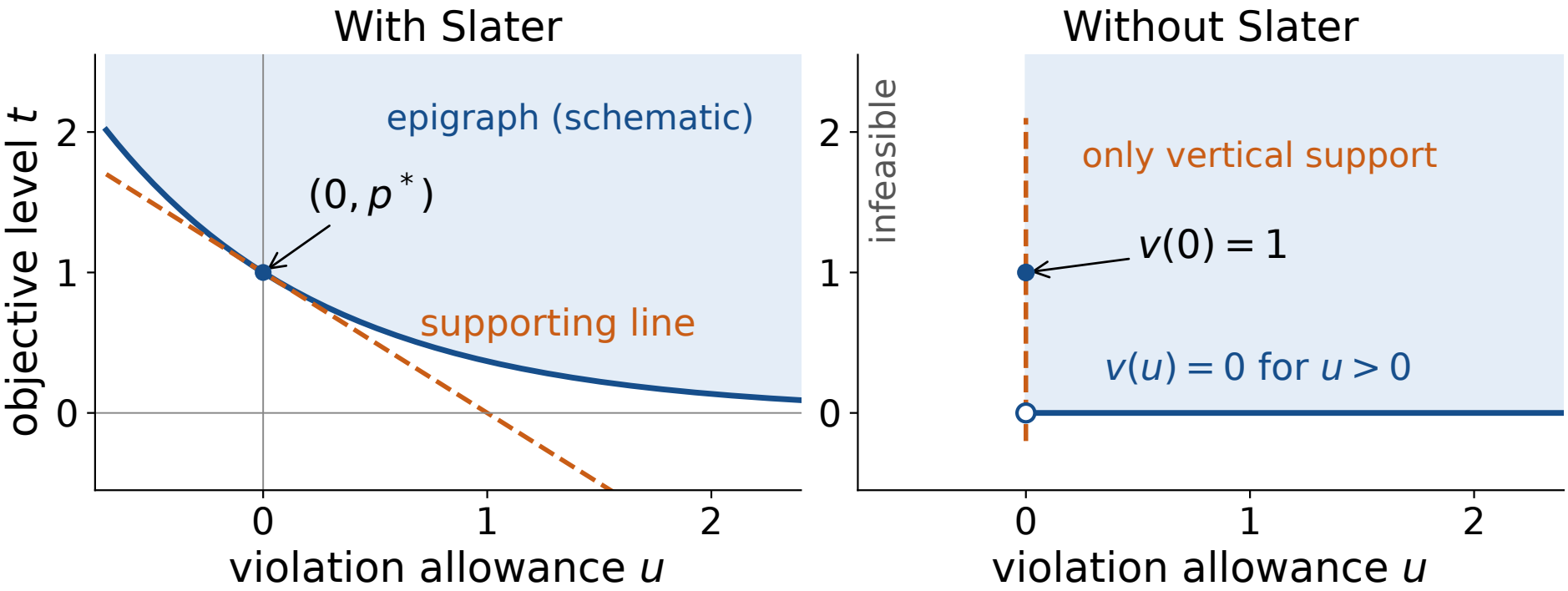
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Illustration

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Without convexity: Slater does not close the gap

$$\min_x (x^2-1)^2 \quad \text{s. t.} \quad x=0, \qquad x^2-1\le 0.$$

At $x=0$, Slater holds, $p^*=1$. But the objective is nonconvex. For $\nu \ge 0$,

$$L(x,\lambda,\nu) = (x^2-1)^2 + \lambda x + \nu(x^2-1).$$

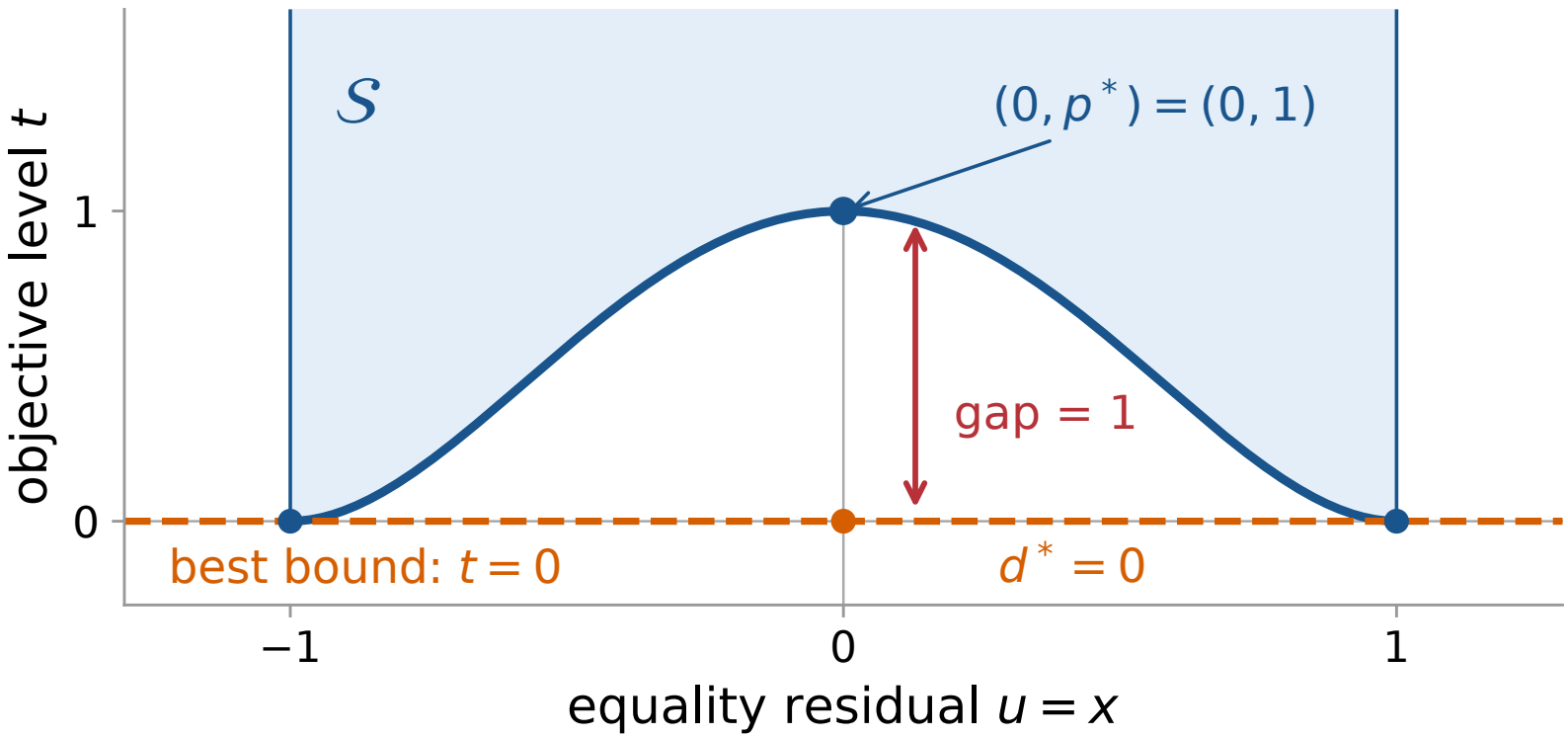
Testing $x=\pm 1$ gives

$$\theta(\lambda,\nu) \le \min\{\lambda,-\lambda\} = -|\lambda| \le 0, \qquad \theta(0,0)=0.$$

$$\boxed{d^* = 0 < 1 = p^*.}$$

The nonconvex example in the (u, t) plane

$$\{(u, t) : |u| \le 1,\ (u^2 - 1)^2 \le t\}.$$



Without Slater: even convex data can leave a gap

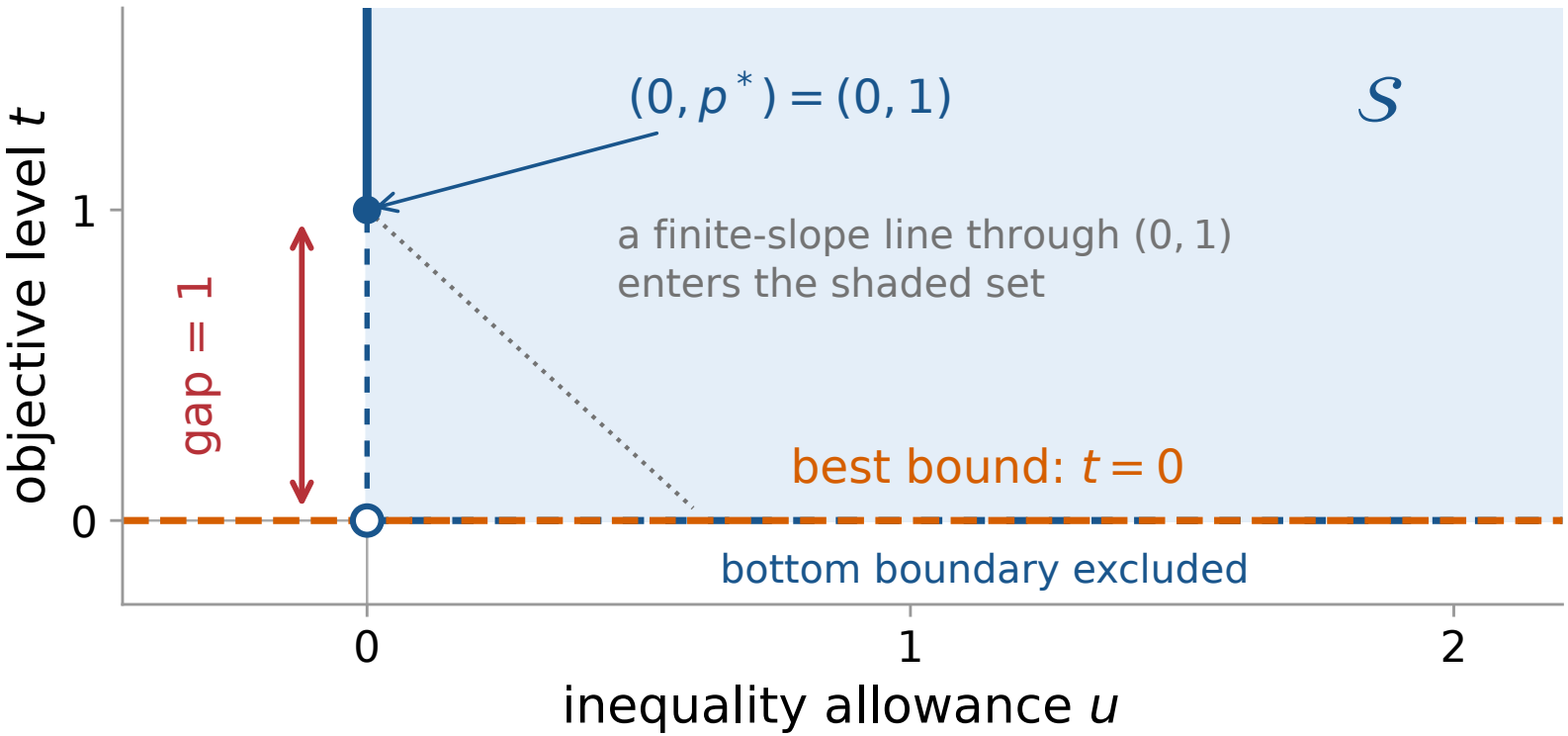
Consider

$$\min_{(x,y)\in\mathbf{R}^2} e^{-x} \quad \text{s. t.} \quad h(x,y) = \left(\sqrt{x^2+y^2} - y\right)^2 \leq 0.$$

- e^{-x} is convex.
- $\sqrt{x^2+y^2} - y$ is convex and nonnegative; hence its square h is convex.
- $h(x,y) \leq 0$ iff $x = 0,\ y \geq 0 \Rightarrow p^* = 1$.
- Slater's condition fails because $h(x,y) \geq 0$.

The convex example in the (u, t) plane

$$\big(\{0\} \times [1, \infty)\big) \cup \big((0, \infty) \times (0, \infty)\big).$$



Neither assumption is necessary for zero gap

Let $f(x) = 1 + 4x^6 - 4x^4$. Consider

$$\min_x f(x) \quad \text{s. t.} \quad x^2 \leq 0.$$

- The objective is nonconvex & Slater fails.
- The only feasible point is $x = 0$, so $p^* = 1$.

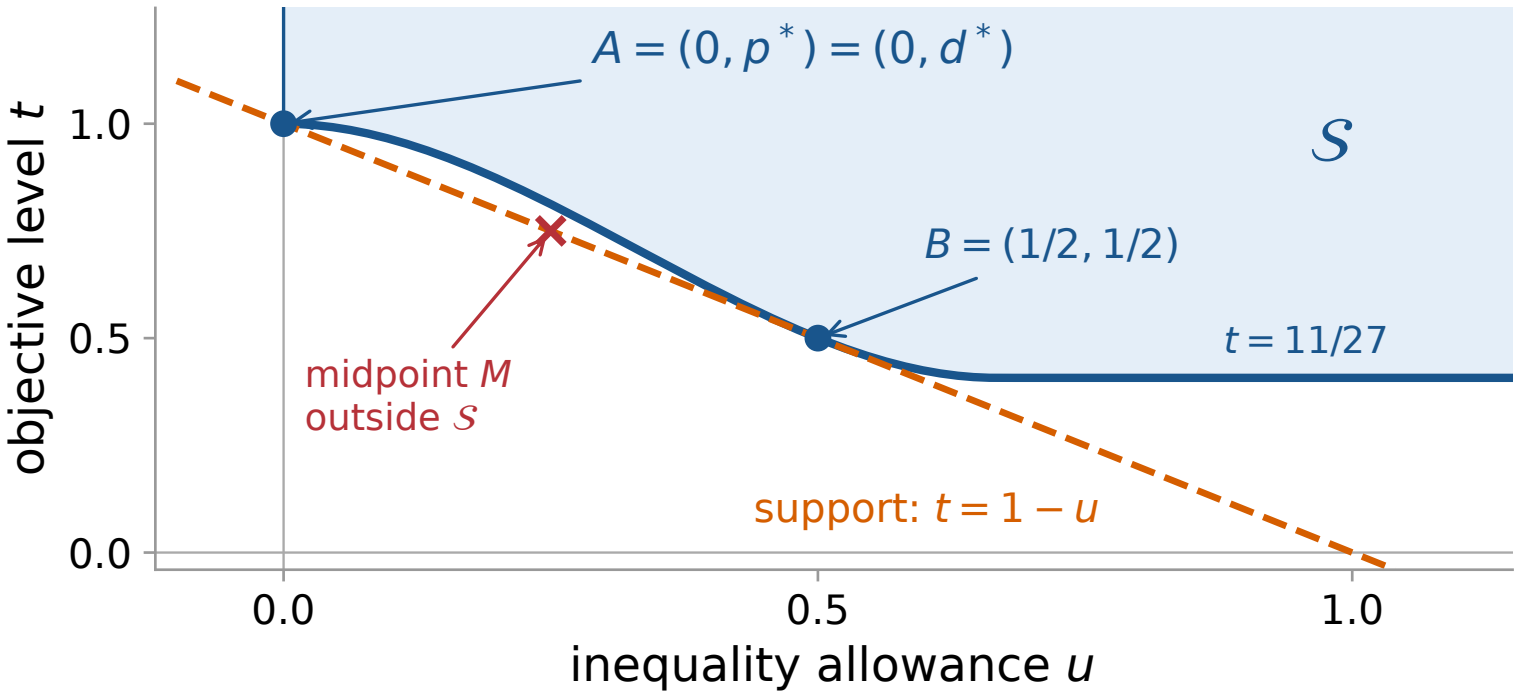
For $\nu = 1$,

$$L(x, 1) = 1 + 4x^6 - 4x^4 + x^2 = 1 + x^2(2x^2 - 1)^2 \geq 1.$$

Equality holds at $x = 0$, so $\theta(1) = 1$. Weak duality certifies $\boxed{d^* = p^* = 1}$.

The zero-gap example in the (u, t) plane

$$\{(u,t) : \exists x,\ x^2 \le u,\ 1 + 4x^6 - 4x^4 \le t\}.$$



Slater guarantees dual attainment, not primal attainment

Consider the convex problem

$$\min_{x \in \mathbf{R}} e^{-x} \quad \text{s. t.} \quad -x \leq 0.$$

Slater holds. On $[0, \infty)$,

$$e^{-x} > 0, \qquad e^{-x} \to 0 \quad (x \to +\infty) \quad \Rightarrow \quad p^* = 0,$$

but no primal optimizer exists.

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$$e^{-x} > 0, \qquad e^{-x} \to 0 \quad (x \to +\infty) \quad \Rightarrow \quad p^* = 0,$$

but no primal optimizer exists. Meanwhile,

$$\theta(\nu) = \inf_x (e^{-x} - \nu x) = \begin{cases} 0, & \nu = 0, \\ -\infty, & \nu > 0. \end{cases}$$

$$\boxed{d^* = p^* = 0, \qquad \nu^* = 0 \text{ is dual optimal.}}$$

Strong duality for linear programming (LP)

$$(\mathrm{P}) \quad \begin{array}{ll} \min\limits_{\mathbf{x}} & \mathbf{c}^T\mathbf{x}, \\ \text{s. t.} & \mathbf{B}\mathbf{x} = \mathbf{b}, \quad \mathbf{x} \geq 0. \end{array} \quad \overset{L(\mathbf{x},\boldsymbol{\lambda},\boldsymbol{\nu})}{\Longleftrightarrow} \quad (\mathrm{D}) \quad \begin{array}{ll} \max\limits_{\boldsymbol{\lambda}} & -\mathbf{b}^T\boldsymbol{\lambda}, \\ \text{s. t.} & \mathbf{c} + \mathbf{B}^T\boldsymbol{\lambda} \geq \mathbf{0}. \end{array}$$

Theorem 19.2 (LP strong duality)

If either "(P) is feasible and $p^* \in \mathbf{R}$" or "(D) is feasible and $d^* \in \mathbf{R}$", then both attain optimal solutions with $p^* = d^*$.

- For affine constraints, Slater's condition is unnecessary.
- If one LP is unbounded, the other is infeasible.
- It is possible that (P) and (D) are infeasible ($p^* = +\infty$, $d^* = -\infty$).

Attained strong duality ⇒ KKT

Theorem 19.3 (Matching values imply KKT)

For differentiable data, if both optima are attained and $p^* = d^* \in \mathbf{R}$, the optimal primal–dual pair satisfies KKT.

Weak duality:

$$d^* = \theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \le L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \le f(\mathbf{x}^*) = p^*.$$

Hence, $p^* = d^*$ gives

- stationarity: $\mathbf{x}^*$ globally minimizes $L(\cdot, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$;
- primal and dual feasibilities naturally hold;
- complementary slackness: $\sum_j \nu_j^* h_j(\mathbf{x}^*) = 0$; each term is nonpositive.

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Conversely?

KKT $\not\Rightarrow$ strong duality

$$\min_x (x^2 - 1)^2 \quad \text{s. t.} \quad x = 0, \quad x^2 - 1 \leq 0.$$

At $(x, \lambda, \nu) = (0, 0, 0)$, KKT holds:

$$\nabla_x L = 4x(x^2 - 1) + \lambda + 2\nu x = 0,$$

primal feasibility, dual signs, and complementarity also hold. Nevertheless,

$$\boxed{d^* = 0 < 1 = p^*.}$$

KKT gives stationarity of L; a matching-value certificate requires global minimization of L.

Convex KKT $\Rightarrow$ attained strong duality

Theorem 19.4 (Convex KKT sufficiency for strong duality)

For convex differentiable f, h_j and affine equalities, any KKT triple $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is primal–dual optimal with zero gap.

Since $\boldsymbol{\nu}^* \geq \mathbf{0}$, $L(\cdot, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ is convex. Thus

$$\nabla_x L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{0} \implies \theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*).$$

Feasibility and complementarity then give

$$\boxed{\theta(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\mathbf{x}^*).}$$

Comparing Slater and KKT: Slater helps obtain multipliers.

The global saddle structure

A triple $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^* \geq \mathbf{0})$ is a global Lagrangian saddle point if

$$\boxed{L(\mathbf{x}^*, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \leq L(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)}$$

for every $\mathbf{x} \in \mathbf{R}^n$, $\boldsymbol{\lambda} \in \mathbf{R}^m$, and $\boldsymbol{\nu} \geq \mathbf{0}$.

- Fix $\mathbf{x} = \mathbf{x}^*$: $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ globally maximizes L.
- Fix $(\boldsymbol{\lambda}, \boldsymbol{\nu}) = (\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$: $\mathbf{x}^*$ globally minimizes L.

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Theorem 19.5 (Saddle points and attained strong duality)

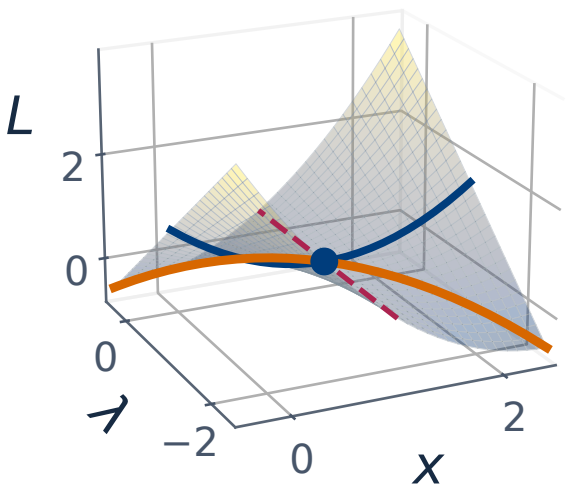
A triple is a global saddle point if and only if it consists of primal and dual optimizers with finite zero gap.

NB. For differentiable data, restricting only $\mathbf{x}$ to a neighborhood of $\mathbf{x}^*$ in the saddle inequalities only implies KKT.

A Lagrangian landscape: the optimum is a saddle

$$\min_x \tfrac{1}{2}x^2 \quad \text{s. t.} \quad x = 1, \qquad L(x,\lambda) = \tfrac{1}{2}x^2 + \lambda(x-1).$$

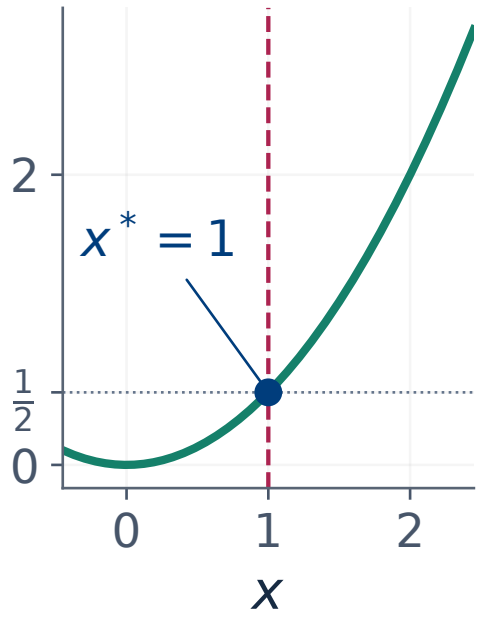
Lagrangian L



$L(x,\lambda^*)$

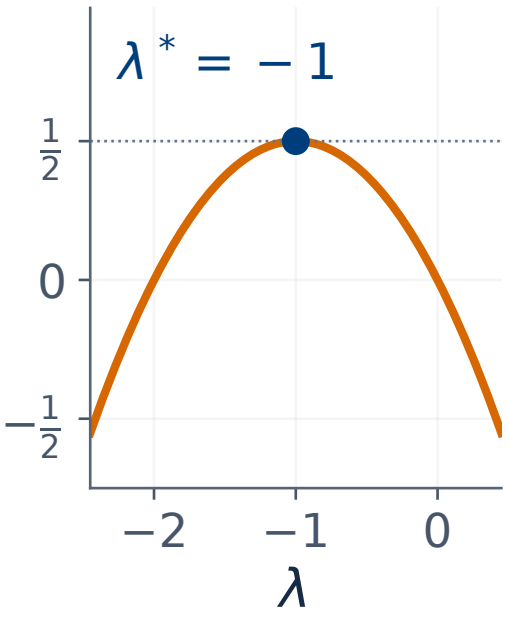
$\theta(\lambda) = L(-\lambda,\lambda)$

Primal f



Feasible: $x = 1$

Dual θ



$p^* = d^* = \frac{1}{2}$

$$x(\lambda) = -\lambda, \quad \theta(\lambda) = -\tfrac{1}{2}\lambda^2 - \lambda, \quad (x^*,\lambda^*) = (1,-1), \quad p^* = d^* = \tfrac{1}{2}.$$

Recall LN: the full Lagrangian Hessian is indefinite

For equality constraints, set $\mathbf{A} = [\nabla g_1, \ldots, \nabla g_m]$ and $\mathbf{H}_L = \nabla^2_{\mathbf{xx}} L$. LN uses

$$\nabla^2_{(\mathbf{x},\boldsymbol{\lambda})} L = \begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix}.$$

In the landscape example,

$$\nabla^2_{(x,\lambda)} L = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \text{eigenvalues } \frac{1 \pm \sqrt{5}}{2}.$$

- Nonzero coupling $\mathbf{A}$ forces the full matrix to be indefinite.
- $\mathbf{H}_L$ is the primal block, not the full Hessian.
- With nonzero coupling, the saddle is not a joint local minimum.

A naive algorithm: one descent step and one ascent step

For equality constraints, one might try simultaneous updates:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta \nabla_x L(\mathbf{x}^{(k)}, \boldsymbol{\lambda}^{(k)}),$$
$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \eta \mathbf{g}(\mathbf{x}^{(k)}).$$

Or alternate, using the new multiplier in the descent step:

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \eta \mathbf{g}(\mathbf{x}^{(k)}),$$
$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} - \eta \nabla_x L(\mathbf{x}^{(k)}, \boldsymbol{\lambda}^{(k+1)}).$$

- These are finite gradient steps, not exact alternating optimization.
- Exact maximization over $\boldsymbol{\lambda}$ is unbounded whenever $\mathbf{g}(\mathbf{x}) \neq \mathbf{0}$.

Even a convex–concave saddle can repel the iterates

Consider $\min_x 0$ subject to $x = 0$. Then

$$L(x, \lambda) = x\lambda, \qquad (x^*, \lambda^*) = (0, 0), \qquad p^* = d^* = 0.$$

The unique saddle exists, but simultaneous updates give

$$x^{(k+1)} = x^{(k)} - \eta\lambda^{(k)}, \qquad \lambda^{(k+1)} = \lambda^{(k)} + \eta x^{(k)}.$$

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Therefore

$$\boxed{(x^{(k+1)})^2 + (\lambda^{(k+1)})^2 = (1 + \eta^2)\big[(x^{(k)})^2 + (\lambda^{(k)})^2\big].}$$

- Every fixed $\eta > 0$ diverges from a nonzero initial point.
- A smaller fixed step only slows the outward motion.

Alternating the steps can still produce persistent motion

For the same $L = x\lambda$, ascent followed by descent gives

$$\lambda^{(k+1)} = \lambda^{(k)} + \eta x^{(k)}, \qquad x^{(k+1)} = x^{(k)} - \eta \lambda^{(k+1)}.$$

$$\begin{pmatrix} x^{(k+1)} \\ \lambda^{(k+1)} \end{pmatrix} = \underbrace{\begin{pmatrix} 1-\eta^2 & -\eta \\ \eta & 1 \end{pmatrix}}_{M} \begin{pmatrix} x^{(k)} \\ \lambda^{(k)} \end{pmatrix}.$$

$$\det M = 1, \qquad \operatorname{tr} M = 2 - \eta^2.$$

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$$\det M = 1, \qquad \operatorname{tr} M = 2 - \eta^2.$$

For $0 < \eta < 2$, both eigenvalues lie on the unit circle: no contraction. At $\eta = 1$, there is a six-cycle:

$$(1,0) \to (0,1) \to (-1,1) \to (-1,0)$$
$$\to (0,-1) \to (1,-1) \to (1,0).$$

The primal–dual coupling generates rotation.

ALM on the same example

$$\min_x \ 0, \quad \text{s. t.} \quad x = 0.$$

Recall the augmented Lagrangian and multiplier correction:

$$L_A(x, \lambda; \rho) = \lambda x + \frac{\rho}{2} x^2, \qquad \rho > 0.$$

Exact inner minimization gives

$$x^{(k+1)} = -\frac{\lambda^{(k)}}{\rho}, \qquad \lambda^{(k+1)} = \lambda^{(k)} + \rho x^{(k+1)} = 0.$$

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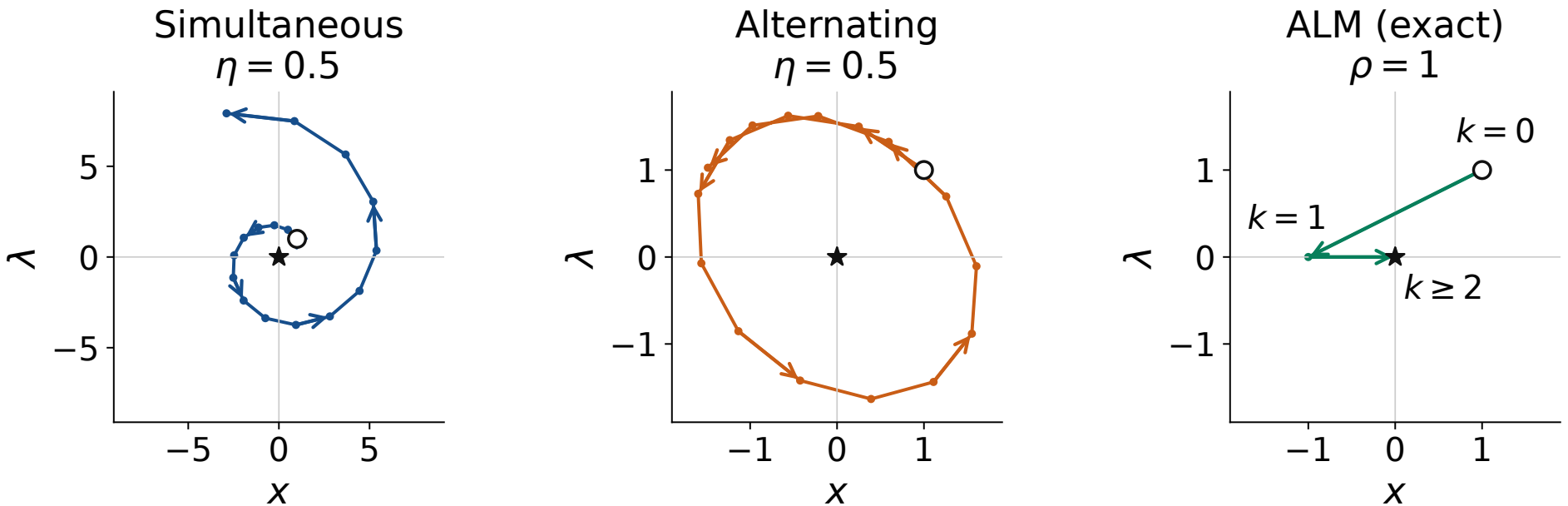
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With $\rho = 1$,

$$\boxed{(x^{(0)}, \lambda^{(0)}) = (1, 1) \ \longrightarrow \ (-1, 0) \ \longrightarrow \ (0, 0).}$$

Three trajectories on one saddle problem

$$L(x, \lambda) = x\lambda, \qquad (x^{(0)}, \lambda^{(0)}) = (1, 1).$$



- Open circle: initial point; star: saddle point. Axes use different scales.
- Gradient methods show 16 steps; ALM: two outer steps.

What did ALM do?

Assume convex differentiable f, affine $\mathbf{g}$.

$$\min_{\mathbf{x}} \; f(\mathbf{x}), \quad \text{s. t.} \;\; \mathbf{g}(\mathbf{x}) = \mathbf{0}.$$

ALM with exact global inner solves and fixed $\rho > 0$:

$$L_A(\mathbf{x}, \boldsymbol{\lambda}; \rho) = L(\mathbf{x}, \boldsymbol{\lambda}) + \frac{\rho}{2}\|\mathbf{g}(\mathbf{x})\|^2.$$

$$\left.\begin{aligned} \mathbf{x}^{(k+1)} &\in \arg\min_{\mathbf{x}} L_A(\mathbf{x}, \boldsymbol{\lambda}^{(k)}; \rho) \\ \boldsymbol{\lambda}^{(k+1)} &= \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)}) \end{aligned}\right\}$$

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$(\mathbf{x}^{(k+1)}, \boldsymbol{\lambda}^{(k+1)})$ is a global saddle point of the regularized Lagrangian

$$\Phi_k(\mathbf{x}, \boldsymbol{\lambda}) := L(\mathbf{x}, \boldsymbol{\lambda}) - \frac{\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^{(k)}\|^2}{2\rho}$$

What did ALM do?

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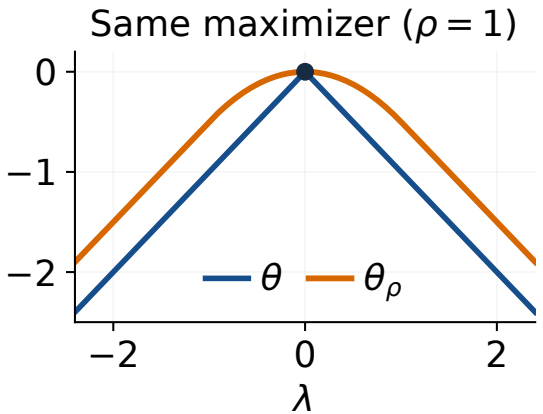
What did ALM do? (Cont'd)

$$\sup_{\boldsymbol{\lambda}} \inf_{\mathbf{x}} \left\{ L(\mathbf{x}, \boldsymbol{\lambda}) - \frac{\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^{(k)}\|^2}{2\rho} \right\} = \sup_{\boldsymbol{\lambda}} \left\{ \theta(\boldsymbol{\lambda}) - \frac{\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^{(k)}\|^2}{2\rho} \right\}.$$

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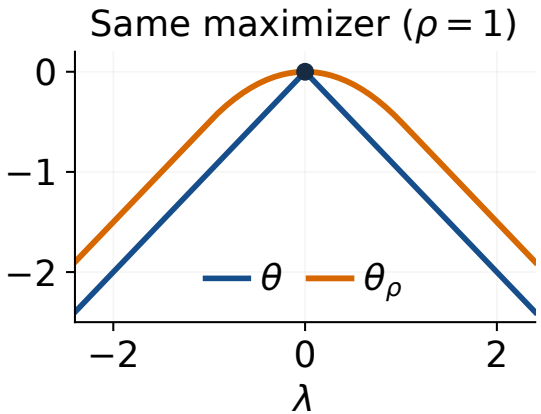
$$\sup_{\boldsymbol{\lambda}} \theta(\boldsymbol{\lambda}) = \sup_{\boldsymbol{\lambda}} \underbrace{\sup_{\mathbf{z}} \left\{ \theta(\mathbf{z}) - \frac{\|\mathbf{z} - \boldsymbol{\lambda}\|^2}{2\rho} \right\}}_{\theta_\rho(\boldsymbol{\lambda})}.$$



What did ALM do? (Cont'd)

$$\sup_{\boldsymbol{\lambda}} \inf_{\mathbf{x}} \left\{ L(\mathbf{x}, \boldsymbol{\lambda}) - \frac{\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^{(k)}\|^2}{2\rho} \right\} = \sup_{\boldsymbol{\lambda}} \left\{ \theta(\boldsymbol{\lambda}) - \frac{\|\boldsymbol{\lambda} - \boldsymbol{\lambda}^{(k)}\|^2}{2\rho} \right\}.$$

$$\sup_{\boldsymbol{\lambda}} \theta(\boldsymbol{\lambda}) = \sup_{\boldsymbol{\lambda}} \underbrace{\sup_{\mathbf{z}} \left\{ \theta(\mathbf{z}) - \frac{\|\mathbf{z} - \boldsymbol{\lambda}\|^2}{2\rho} \right\}}_{\theta_\rho(\boldsymbol{\lambda})}.$$



The proximal term smoothes out the nonsmoothness:

$$\nabla \theta_\rho(\boldsymbol{\lambda}^{(k)}) = \mathbf{g}(\mathbf{x}^{(k+1)}).$$

Gradient ascent on θ_ρ with the step size ρ:

$$\boldsymbol{\lambda}^{(k+1)} = \boldsymbol{\lambda}^{(k)} + \rho \nabla \theta_\rho(\boldsymbol{\lambda}^{(k)}) = \boldsymbol{\lambda}^{(k)} + \rho \mathbf{g}(\mathbf{x}^{(k+1)}).$$

Take-home messages

- Sufficient conditions for strong duality: convexity + Slater/KKT; LP.
- Strong duality does not come with attainment.
- Attained zero gap $\Leftrightarrow$ global Lagrangian saddle.
- Naive gradient descent–ascent can fail even when close to saddle.
- Convex-affine ALM with exact inner solves: gradient ascent on the smoothed dual.

"Last" lecture: "gradient" ascent on the original dual function.